

Dose- and gender-dependent interactions between coffee consumption and serum GGT activity in alcohol consumers.

AIMS: Coffee consumption has been recently linked with decreased blood gamma-glutamyltransferase (GGT) activities and protection from alcoholic liver disease. To explore the relationship and dose response, we assessed the impacts of coffee and alcohol intake on serum GGT activity in apparently healthy men and women with varying levels of coffee and alcohol consumption. **METHODS:** Data on coffee, alcohol consumption and serum GGT activities were collected from 18,899 individuals (8807 men and 10,092 women), mean age 48 years, range 25-74 years, who participated in a large national cross-sectional health survey. Body mass index, smoking index and age were used as covariates in all analyses. **RESULTS:** Among the study population, 89.8% reported varying levels of coffee consumption; 6.9% were abstainers from alcohol, 86.1% moderate drinkers, 3.7% heavy drinkers and 3.3% former drinkers. In men, the elevation of GGT induced by heavy drinking (>280 g/week) was found to be significantly reduced by coffee consumption exceeding 4 cups per day. A similar trend was also observed among women, which however, did not reach statistical significance. **CONCLUSION:** Coffee modulates the effect of ethanol on serum GGT activities in a dose- and gender-dependent manner. These observations should be implicated in studies on the possible hepatoprotective effects of coffee in alcohol consumers.